National Center for Functional Glycomics

Project Number
3P41GM103694-06S1

Contact PI/Project Leader
CUMMINGS, RICHARD D

Awardee Organization
BETH ISRAEL DEACONESS MEDICAL CENTER

Description

Abstract Text

DESCRIPTION (provided by applicant): Here we propose to establish a "National Center for Functional Glycomics" (NCFG) that is focused on 5 major objectives; Technology Research and Development (TR&D); Driving Biomedical Projects (DBPs) tied to TR&D Projects; Collaborative Services; Training; and Dissemination of Technologies and Research Discoveries. The development of this application is an outgrowth of the very successful leadership provided by the Glycomics Center at Emory University School of Medicine, which housed the Protein-Glycan Interaction Core (Core H) of the successful Consortium for Functional Glycomics (CFG), a unique NIGMS-funded resource from 2001-2011 that has been utilized by hundreds of investigators worldwide. The Center's applications of the CFG's Defined Glycan Microarray, which has received >800 requests for analyses in the past 8 years, has revolutionized studies on protein-glycan interactions. Defined glycan microarrays have become the benchmark in studying the glycan-binding specificity of lectins, antibodies, and viruses, which for this grant application are termed glycan-binding proteins (GBPs). The value of immobilized glycans in studying GBPs and functional glycomics is evidenced by the collaborations already established in the Center with over 600 Principal Investigators worldwide, resulting in >350 peer-reviewed publications in the past 8 years. Over the past several years, the members of the Glycomics Center at Emory, who are the lead investigators in this proposed NCFG, have provided innovative leadership in developing technologies for studying the functions of cellular glycomes and for glycomic analyses. In the 3 TR&D Projects proposed for this NCFG, we will 1) expand glycan microarray technologies, 2) develop "Shotgun Glycomics" as a general method for studying natural cell-derived glycan recognition, and 3) develop different glycan display technologies, as well as cellular avatars - particles that represent cellular glycomes in unique presentations. The DBPs include studies on the roles of glycans in microbial and viral infections, glycans and GBPs in cell adhesion, innate and adaptive immune responses, and recognition factors within the glycocalyx that contribute to GBP interactions. The NCFG will use these advances to drive a strong Collaborative Service component to aid the biomedical research community in the using and developing of glycan microarrays and other approaches to explore glycan recognition. We will offer a robust Training program, as well as initiate multiple venues and strategies to Disseminate the technological and scientific developments in the NCFG.

Public Health Relevance Statement

PUBLIC HEALTH RELEVANCE: This proposed National Center for Functional Glycomics (NCFG) will provide a unique technological and biological resource for studying the structure and function of human and animal glycomes, through innovative glycan microarray technologies, and novel glycan presentations, as well as explore an important set of driving biological projects that exploit the technology developments. The NCFG will provide a unique national resource in collaboration, training, and dissemination of results in the field that will elevate and propel important advances in glycomics.

NIH Spending Category

Biotechnology

Project Terms

Address | Animals | Antibodies | Applications Grants | Area | Automobile Driving | Beds | Benchmarking | Binding | Binding Proteins | Biological | Biomedical Research | Biomedical Technology | Cell Adhesion | Cell physiology | Cells | Collaborations | Communities | Data | Development | Fostering | Funding

Glycocalyx | Human | Immobilization | Information Dissemination | Lead | Leadership | Lectin | Methods | Microarray Analysis | National Institute of General Medical Sciences | Peer Review | Polysaccharides | Principal Investigator | Proteins | Publications | Research | Research Personnel | Resources | Role | Services | Shotguns | Specificity | Structure | Technology | Testing | Training | Training Programs | Universities | Virus | Virus Diseases | adaptive immune response | innovation | medical schools | member | microbial | new technology | novel | particle | public health relevance | synergism

Details

Contact PI/ Project Leader

Name
CUMMINGS, RICHARD D

Title
PROFESSOR OF SURGERY

Contact
View Email

Other PIs

Not Applicable

Program Official

Name
SAMMAK, PAUL J

Contact
View Email

Organization

Name
BETH ISRAEL DEACONESS MEDICAL CENTER

City
BOSTON

Country
UNITED STATES (US)

Department Type
Unavailable

Organization Type
Independent Hospitals

State Code
MA

Congressional District
07

Other Information

Opportunity Number
PAR-10-225

Study Section
Special Emphasis Panel[ZRG1-BCMB-K(40)P]

Fiscal Year
2018

Award Notice Date
24-August-2018

Administering Institutes or Centers
National Institute of General Medical Sciences

CFDA Code
859

DUNS Number
071723621

UEI
C1CPANL3EWK4

Project Start Date **01-August-2013**

Project End Date **30-June-2020**

Budget Start Date **01-July-2017**

Budget End Date **30-June-2020**

Project Funding Information for 2018

Total Funding
$1,163,832

Direct Costs
$665,047

Indirect Costs
$498,785

Year	Funding IC	FY Total Cost	FY Total Cost by IC
		stitute of General Medical Sciences	$1,163,832

NIH Categorical Spending

Click here for more information on NIH Categorical Spending

Funding IC	FY Total Cost by IC	NIH Spending Category
NATIONAL INSTITUTE OF GENERAL MEDICAL SCIENCES	$1,163,832	Biotechnology

Sub Projects

No Sub Projects information available for 3P41GM103694-06S1

Publications

Export | Disclaimer

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR	tio	Similar Publications	CitedBy	iCite RCR
New carbohydrate binding domains identified by phage display based functional metagenomic screens of human gut microbiota.									
Communications biology 2023 Apr 05; 6 (1) 371	Akhtar, Akil; Lata, Madhu; Sunsunwal, Sonali; Yadav, Amit; Lnu, Kajal; Subramanian, Srikrishna; Ramya, T N C	2023							
Chemokine binding to PSGL-1 is controlled by O-glycosylation and tyrosine sulfation.									
Cell chemical biology 2023 Aug 17; 30 (8) 893-905.e7	Goth, Christoffer K; Mehta, Akul Y; McQuillan, Alyssa M; Baker, Kelly J; Hanes, Melinda S; Park, Simon S; Stavenhagen, Kathrin; Hjortø, Gertrud M; Heimburg-Molinaro, Jamie; Chaikof, Elliot L; Rosenkilde, Mette M; Cummings, Richard D View Less	2023							
Structure-function correlates of fibrinogen binding by Acinetobacter adhesins critical in catheter-associated urinary tract infections.									
Proceedings of the National Academy of Sciences of the United States of America 2023 Jan 24; 120 (4) e2212694120	Tamadonfar, Kevin O; Di Venanzio, Gisela; Pinkner, Jerome S; Dodson, Karen W; Kalas, Vasilios; Zimmerman, Maxwell I; Bazan Villicana, Jesus; Bowman, Gregory R; Feldman, Mario F; Hultgren, Scott J View Less	2023							
Sialylation regulates neutrophil transepithelial migration, CD11b/CD18 activation, and intestinal mucosal inflammatory function.									
JCI insight 2023 Mar 08; 8 (5)	Azcutia, Veronica; Kelm, Matthias; Fink, Dylan; Cummings, Richard D; Nusrat, Asma; Parkos, Charles A; Brazil, Jennifer C	2023							
Targeting altered glycosylation in secreted tumor glycoproteins for broad cancer detection.									
Glycobiology 2023 Aug 14; 33 (7) 567-578	Kudelka, Matthew R; Gu, Wei; Matsumoto, Yasuyuki; Ju, Tongzhong; Barnes Ii, Richard H; Kardish, Robert J; Heimburg-Molinaro, Jamie; Lehoux, Sylvain; Zeng, Junwei; Cohen, Cynthia; Robinson, Brian S; Shah, Kinjal S; Chaikof, Elliot L; Stowell, Sean R; Cummings, Richard D View Less	2023							
Increased Galectin-9 Levels Correlate with Disease Activity in Patients with DMARD-Naïve Rheumatoid Arthritis and Modulate the Secretion of MCP-1 and IL-6 from Synovial Fibroblasts.									
Cells 2023 Jan 15; 12 (2)	Nielsen, Morten A; Køster, Ditte; Mehta, Akul Y; Stengaard-Pedersen, Kristian; Busson, Pierre; Junker, Peter; Hørslev-Petersen, Kim; Hetland, Merete Lund; Østergaard, Mikkel; Hvid, Malene; Leffler, Hakon; Kragstrup, Tue W; Cummings, Richard D; Deleuran, Bent View Less	2023							

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Anti-schistosomal immunity to core xylose/fucose in N-glycans.					
Frontiers in molecular biosciences 2023; 10 1142620	Prasanphanich, Nina Salinger; Leon, Kristoffer; Secor, W Evan; Shoemaker, Charles B; Heimburg-Molinaro, Jamie; Cummings, Richard D	2023			
Mammalian brain glycoproteins exhibit diminished glycan complexity compared to other tissues.					
Nature communications 2022 Jan 12; 13 (1) 275	Williams, Sarah E; Noel, Maxence; Lehoux, Sylvain; Cetinbas, Murat; Xavier, Ramnik J; Sadreyev, Ruslan I; Scolnick, Edward M; Smoller, Jordan W; Cummings, Richard D; Mealer, Robert G View Less	2022			iCite 6.29
Unraveling function and diversity of bacterial lectins in the human microbiome.					
Nature communications 2022 Jun 03; 13 (1) 3101	Cohen, Louis J; Han, Sun M; Lau, Pearson; Guisado, Daniela; Liang, Yupu; Nakashige, Toshiki G; Ali, Thamina; Chiang, David; Rahman, Adeeb; Brady, Sean F View Less	2022			
Sialylated and sulfated N-Glycans in MDCK and engineered MDCK cells for influenza virus studies.					
Scientific reports 2022 Jul 26; 12 (1) 12757	Byrd-Leotis, Lauren; Jia, Nan; Matsumoto, Yasuyuki; Lu, Dongli; Kawaoka, Yoshihiro; Steinhauer, David A; Cummings, Richard D	2022			
Use of a glycomics array to establish the anti-carbohydrate antibody repertoire in type 1 diabetes.					
Nature communications 2022 Nov 01; 13 (1) 6527	Tran, Paul M H; Dong, Fran; Kim, Eileen; Richardson, Katherine P; Tran, Lynn K H; Waugh, Kathleen; Hopkins, Diane; Cummings, Richard D; Wang, Peng George; Rewers, Marian J; She, Jin-Xiong; Purohit, Sharad View Less	2022			
Structural analysis of N-glycans in chicken trachea and lung reveals potential receptors of chicken influenza viruses.					
Scientific reports 2022 Feb 08; 12 (1) 2081	Suzuki, Noriko; Abe, Tatsuya; Natsuka, Shunji	2022			iCite 1.77
The schizophrenia-associated variant in SLC39A8 alters protein glycosylation in the mouse brain.					
Molecular psychiatry 2022 Mar; 27 (3) 1405-1415	Mealer, Robert G; Williams, Sarah E; Noel, Maxence; Yang, Bo; D'Souza, Alexandria K; Nakata, Toru; Graham, Daniel B; Creasey, Elizabeth A; Cetinbas, Murat; Sadreyev, Ruslan I; Scolnick, Edward M; Woo, Christina M; Smoller, Jordan W; Xavier, Ramnik J; Cummings, Richard D View Less	2022			iCite 2.57
N-glycosylation of mannose receptor (CD206) regulates glycan binding by C-type lectin domains.					
The Journal of biological chemistry 2022 Dec; 298 (12) 102591	Stavenhagen, Kathrin; Mehta, Akul Y; Laan, Lisa; Gao, Chao; Heimburg-Molinaro, Jamie; van Die, Irma; Cummings, Richard D	2022			
Mucin O-glycans are natural inhibitors of Candida albicans pathogenicity.					

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Nature chemical biology 2022 Jul; 18 (7) 762-773	Takagi, Julie; Aoki, Kazuhiro; Turner, Bradley S; Lamont, Sabrina; Lehoux, Sylvain; Kavanaugh, Nicole; Gulati, Megha; Valle Arevalo, Ashley; Lawrence, Travis J; Kim, Colin Y; Bakshi, Bhavya; Ishihara, Mayumi; Nobile, Clarissa J; Cummings, Richard D; Wozniak, Daniel J; Tiemeyer, Michael; Hevey, Rachel; Ribbeck, Katharina View Less	2022			iCite 3.54

Computationally guided conversion of the specificity of E-selectin to mimic that of Siglec-8.

Proceedings of the National Academy of Sciences of the United States of America 2022 Oct 11; 119 (41) e2117743119	Wang, Xiaocong; Hanes, Melinda S; Cummings, Richard D; Woods, Robert J	2022			

Identification of Glycan-Specific Variable Lymphocyte Receptors Using Yeast Surface Display and Glycan Microarrays.

Methods in molecular biology (Clifton, N.J.) 2022; 2421 73-89	McKitrick, Tanya R; Hanes, Melinda S; Rosenberg, Charles S; Heimburg-Molinaro, Jamie; Cooper, Max D; Herrin, Brantley R; Cummings, Richard D	2022			

A Useful Guide to Lectin Binding: Machine-Learning Directed Annotation of 57 Unique Lectin Specificities.

ACS chemical biology 2022 Nov 18; 17 (11) 2993-3012	Bojar, Daniel; Meche, Lawrence; Meng, Guanmin; Eng, William; Smith, David F; Cummings, Richard D; Mahal, Lara K	2022			iCite 15.21

Identification of lamprey variable lymphocyte receptors that target the brain vasculature.

Scientific reports 2022 Apr 11; 12 (1) 6044	Lajoie, Jason M; Katt, Moriah E; Waters, Elizabeth A; Herrin, Brantley R; Shusta, Eric V	2022			

Origins of glycan selectivity in streptococcal Siglec-like adhesins suggest mechanisms of receptor adaptation.

Nature communications 2022 May 18; 13 (1) 2753	Bensing, Barbara A; Stubbs, Haley E; Agarwal, Rupesh; Yamakawa, Izumi; Luong, Kelvin; Solakyildirim, Kemal; Yu, Hai; Hadadianpour, Azadeh; Castro, Manuel A; Fialkowski, Kevin P; Morrison, KeAndreya M; Wawrzak, Zdzislaw; Chen, Xi; Lebrilla, Carlito B; Baudry, Jerome; Smith, Jeremy C; Sullam, Paul M; Iverson, T M View Less	2022			

SARS-CoV-2 and other coronaviruses bind to phosphorylated glycans from the human lung.

Virology 2021 10; 562 142-148	Byrd-Leotis, Lauren; Lasanajak, Yi; Bowen, Thomas; Baker, Kelly; Song, Xuezheng; Suthar, Mehul S; Cummings, Richard D; Steinhauer, David A	2021			iCite 1.40

The lytic polysaccharide monooxygenase CbpD promotes Pseudomonas aeruginosa virulence in systemic infection.

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Nature communications 2021 02 23; 12 (1) 1230	Askarian, Fatemeh; Uchiyama, Satoshi; Masson, Helen; Sørensen, Henrik Vinther; Golten, Ole; Bunæs, Anne Cathrine; Mekasha, Sophanit; Røhr, Åsmund Kjendseth; Kommedal, Eirik; Ludviksen, Judith Anita; Arntzen, Magnus Ø; Schmidt, Benjamin; Zurich, Raymond H; van Sorge, Nina M; Eijsink, Vincent G H; Krengel, Ute; Mollnes, Tom Eirik; Lewis, Nathan E; Nizet, Victor; Vaaje-Kolstad, Gustav View Less	2021			6.68

Tumor cells express pauci- and oligomannosidic N-glycans in glycoproteins recognized by the mannose receptor (CD206).

Cellular and molecular life sciences : CMLS 2021 Jul; 78 (14) 5569-5585	Stavenhagen, Kathrin; Laan, Lisa C; Gao, Chao; Mehta, Akul Y; Heimburg-Molinaro, Jamie; Glickman, Jonathan N; van Die, Irma; Cummings, Richard D	2021			1.27

Mucin Glycans Signal through the Sensor Kinase RetS to Inhibit Virulence-Associated Traits in Pseudomonas aeruginosa.

Current biology : CB 2021 01 11; 31 (1) 90-102.e7	Wang, Benjamin X; Wheeler, Kelsey M; Cady, Kyle C; Lehoux, Sylvain; Cummings, Richard D; Laub, Michael T; Ribbeck, Katharina	2021			5.01

Resident and elicited murine macrophages differ in expression of their glycomes and glycan-binding proteins.

Cell chemical biology 2021 Apr 15; 28 (4) 567-582.e4	Park, Diane D; Chen, Jiaxuan; Kudelka, Matthew R; Jia, Nan; Haller, Carolyn A; Kosaraju, Revanth; Premji, Alykhan M; Galizzi, Melina; Nairn, Alison V; Moremen, Kelley W; Cummings, Richard D; Chaikof, Elliot L View Less	2021			0.77

Human group A rotavirus P[25] VP8* specifically binds to A-type histo-blood group antigen.

Virology 2021 03; 555 56-63	Li, Dandi; Wang, Mengxuan; Qi, Jianxun; Zhang, Qing; Wang, Hong; Pang, Lili; Sun, Xiaoman; Duan, Zhaojun	2021			1.30

Proteomic analysis of the host-pathogen interface in experimental cholera.

Nature chemical biology 2021 Nov; 17 (11) 1199-1208	Zoued, Abdelrahim; Zhang, Hailong; Zhang, Ting; Giorgio, Rachel T; Kuehl, Carole J; Fakoya, Bolutife; Sit, Brandon; Waldor, Matthew K	2021			0.90

mAb Das-1 recognizes 3'-Sulfated Lewis A/C, which is aberrantly expressed during metaplastic and oncogenic transformation of several gastrointestinal Epithelia.

PloS one 2021; 16 (12) e0261082	Brown, Jeffrey W; Das, Koushik K; Kalas, Vasilios; Das, Kiron M; Mills, Jason C	2021			0.61

Different glycoforms of alpha-1-acid glycoprotein contribute to its functional alterations in platelets and neutrophils.

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Journal of leukocyte biology 2021 05; 109 (5) 915-930	Sumanth, Mosale Seetharam; Jacob, Shancy P; Abhilasha, Kandahalli Venkataranganayaka; Manne, Bhanu Kanth; Basrur, Venkatesha; Lehoux, Sylvain; Campbell, Robert A; Yost, Christian C; McIntyre, Thomas M; Cummings, Richard D; Weyrich, Andrew S; Rondina, Matthew T; Marathe, Gopal K View Less	2021			iCite 0.43
The Functional Characterization of Bat and Human P[3] Rotavirus VP8*s.					
Virologica Sinica 2021 Oct; 36 (5) 1187-1196	Li, Dandi; Wang, Mengxuan; Mao, Tongyao; Wang, Mingwen; Zhang, Qing; Wang, Hong; Pang, Lili; Sun, Xiaoman; Duan, Zhaojun	2021			iCite 0.36
Differential recognition of oligomannose isomers by glycan-binding proteins involved in innate and adaptive immunity.					
Science advances 2021 06; 7 (24)	Gao, Chao; Stavenhagen, Kathrin; Eckmair, Barbara; McKitrick, Tanya R; Mehta, Akul Y; Matsumoto, Yasuyuki; McQuillan, Alyssa M; Hanes, Melinda S; Eris, Deniz; Baker, Kelly J; Jia, Nan; Wei, Mohui; Heimburg-Molinaro, Jamie; Ernst, Beat; Cummings, Richard D View Less	2021			iCite 1.84
Novel lamprey antibody recognizes terminal sulfated galactose epitopes on mammalian glycoproteins.					
Communications biology 2021 06 03; 4 (1) 674	McKitrick, Tanya R; Bernard, Steffen M; Noll, Alexander J; Collins, Bernard C; Goth, Christoffer K; McQuillan, Alyssa M; Heimburg-Molinaro, Jamie; Herrin, Brantley R; Wilson, Ian A; Cooper, Max D; Cummings, Richard D View Less	2021			iCite 1.74
Major differences in glycosylation and fucosyltransferase expression in low-grade versus high-grade bladder cancer cell lines.					
Glycobiology 2021 12 18; 31 (11) 1444-1463	Ezeabikwa, Bernadette; Mondal, Nandini; Antonopoulos, Aristotelis; Haslam, Stuart M; Matsumoto, Yasuyuki; Martin-Caraballo, Miguel; Lehoux, Sylvain; Mandalasi, Msano; Ishaque, Ali; Heimburg-Molinaro, Jamie; Cummings, Richard D; Nyame, Anthony K View Less	2021			iCite 0.91
Identification of Tn antigen O-GalNAc-expressing glycoproteins in human carcinomas using novel anti-Tn recombinant antibodies.					
Glycobiology 2020 04 20; 30 (5) 282-300	Matsumoto, Yasuyuki; Kudelka, Matthew R; Hanes, Melinda S; Lehoux, Sylvain; Dutta, Sucharita; Jones, Mark B; Stackhouse, Kathryn A; Cervoni, Gabrielle E; Heimburg-Molinaro, Jamie; Smith, David F; Ju, Tongzhong; Chaikof, Elliot L; Cummings, Richard D View Less	2020			iCite 1.99
Structure and Ligand-Binding Properties of the O Antigen ABC Transporter Carbohydrate-Binding Domain.					

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
Structure (London, England : 1993) 2020 02 04; 28 (2) 252-258.e2	Bi, Yunchen; Zimmer, Jochen	2020			0.72
Emerging patterns of tyrosine sulfation and O-glycosylation cross-talk and co-localization.					
Current opinion in structural biology 2020 06; 62 102-111	Mehta, Akul Y; Heimburg-Molinaro, Jamie; Cummings, Richard D; Goth, Christoffer K	2020			1.57
GlycoGlyph: a glycan visualizing, drawing and naming application.					
Bioinformatics (Oxford, England) 2020 06 01; 36 (11) 3613-3614	Mehta, Akul Y; Cummings, Richard D	2020			2.41
O-glycan recognition and function in mice and human cancers.					
The Biochemical journal 2020 04 30; 477 (8) 1541-1564	Cervoni, Gabrielle E; Cheng, Jane J; Stackhouse, Kathryn A; Heimburg-Molinaro, Jamie; Cummings, Richard D	2020			3.79
Enhanced Pro-apoptotic Effects of Fe(II)-Modified IVIG on Human Neutrophils.					
Frontiers in immunology 2020; 11 973	Graeter, Stefanie; Schneider, Christoph; Verschoor, Daniëlle; von Däniken, Sandro; Seibold, Frank; Yawalkar, Nikhil; Villiger, Peter; Dimitrov, Jordan D; Smith, David F; Cummings, Richard D; Simon, Hans-Uwe; Vassilev, Tchavdar; von Gunten, Stephan View Less	2020			0.32
Parallel Glyco-SPOT Synthesis of Glycopeptide Libraries.					
Cell chemical biology 2020 09 17; 27 (9) 1207-1219.e9	Mehta, Akul Y; Veeraiah, Ravi Kumar H; Dutta, Sucharita; Goth, Christoffer K; Hanes, Melinda S; Gao, Chao; Stavenhagen, Kathrin; Kardish, Robert; Matsumoto, Yasuyuki; Heimburg-Molinaro, Jamie; Boyce, Michael; Pohl, Nicola L B; Cummings, Richard D View Less	2020			0.81
Intestinal epithelial glycosylation in homeostasis and gut microbiota interactions in IBD.					
Nature reviews. Gastroenterology & hepatology 2020 Oct; 17 (10) 597-617	Kudelka, Matthew R; Stowell, Sean R; Cummings, Richard D; Neish, Andrew S	2020			7.00
The Glycoprotease CpaA Secreted by Medically Relevant Acinetobacter Species Targets Multiple O-Linked Host Glycoproteins.					
mBio 2020 10 06; 11 (5)	Haurat, M Florencia; Scott, Nichollas E; Di Venanzio, Gisela; Lopez, Juvenal; Pluvinage, Benjamin; Boraston, Alisdair B; Ferracane, Michael J; Feldman, Mario F View Less	2020			2.23
Tools for generating and analyzing glycan microarray data.					
Beilstein journal of organic chemistry 2020; 16 2260-2271	Mehta, Akul Y; Heimburg-Molinaro, Jamie; Cummings, Richard D	2020			0.90
PDX-derived organoids model in vivo drug response and secrete biomarkers.					

Journal (Link to PubMed abstract)	Authors	Publication Year	Similar Publications	CitedBy	iCite RCR
JCI insight 2020 11 05; 5 (21)	Huang, Ling; Bockorny, Bruno; Paul, Indranil; Akshinthala, Dipikaa; Frappart, Pierre-Oliver; Gandarilla, Omar; Bose, Arindam; Sanchez-Gonzalez, Veronica; Rouse, Emily E; Lehoux, Sylvain D; Pandell, Nicole; Lim, Christine M; Clohessy, John G; Grossman, Joseph; Gonzalez, Raul; Del Pino, Sofia Perea; Daaboul, George; Sawhney, Mandeep S; Freedman, Steven D; Kleger, Alexander; Cummings, Richard D; Emili, Andrew; Muthuswamy, Lakshmi B; Hidalgo, Manuel; Muthuswamy, Senthil K View Less	2020			iCite 3.89

GlyMDB: Glycan Microarray Database and analysis toolset.

Bioinformatics (Oxford, England) 2020 04 15; 36 (8) 2438-2442	Cao, Yiwei; Park, Sang-Jun; Mehta, Akul Y; Cummings, Richard D; Im, Wonpil	2020			iCite 1.08

Regulation of neutrophil function by selective targeting of glycan epitopes expressed on the integrin CD11b/CD18.

FASEB journal : official publication of the Federation of American Societies for Experimental Biology 2020 02; 34 (2) 2326-2343	Kelm, Matthias; Lehoux, Sylvain; Azcutia, Veronica; Cummings, Richard D; Nusrat, Asma; Parkos, Charles A; Brazil, Jennifer C	2020			iCite 1.68

Targeting epithelium-expressed sialyl Lewis glycans improves colonic mucosal wound healing and protects against colitis.

JCI insight 2020 06 18; 5 (12)	Kelm, Matthias; Quiros, Miguel; Azcutia, Veronica; Boerner, Kevin; Cummings, Richard D; Nusrat, Asma; Brazil, Jennifer C; Parkos, Charles A	2020			iCite 1.22

Unique repertoire of anti-carbohydrate antibodies in individual human serum.

Scientific reports 2020 09 22; 10 (1) 15436	Luetscher, Ralph N D; McKitrick, Tanya R; Gao, Chao; Mehta, Akul Y; McQuillan, Alyssa M; Kardish, Robert; Boligan, Kayluz Frias; Song, Xuezheng; Lu, Lenette; Heimburg-Molinaro, Jamie; von Gunten, Stephan; Alter, Galit; Cummings, Richard D View Less	2020			iCite 1.17

Development of smart anti-glycan reagents using immunized lampreys.

Communications biology 2020 02 28; 3 (1) 91	McKitrick, Tanya R; Goth, Christoffer K; Rosenberg, Charles S; Nakahara, Hirotomo; Heimburg-Molinaro, Jamie; McQuillan, Alyssa M; Falco, Rosalia; Rivers, Nicholas J; Herrin, Brantley R; Cooper, Max D; Cummings, Richard D View Less	2020			iCite 2.08

Better survival is observed in cervical cancer patients positive for specific anti-glycan antibodies and receiving brachytherapy.

Gynecologic oncology 2020 04; 157 (1) 181-187	Purohit, Sharad; Ferris, Daron G; Alvarez, Manual; Tran, Paul M H; Tran, Lynn K H; Mysona, David P; Hopkins, Diane; Zhi, Wenbo; Dun, Boying; Wallbillich, John J; Cummings, Richard D; Wang, Peng George; She, Jin-Xiong View Less	2020			iCite 0.54

Patents

No Patents information available for 3P41GM103694-06S1

Outcomes

The Project Outcomes shown here are displayed verbatim as submitted by the Principal Investigator (PI) for this award. Any opinions, findings, and conclusions or recommendations expressed are those of the PI and do not necessarily reflect the views of the National Institutes of Health. NIH has not endorsed the content below.

No Outcomes available for 3P41GM103694-06S1

Clinical Studies

No Clinical Studies information available for 3P41GM103694-06S1

News and More

Related News Releases

News	Journal Article	PubMed Abstract	Release Date
[illegible] Bacteria			May 2014

Related NIH Research Matters

News	PubMed Abstract	Release Date
[illegible] Typhoid Toxin		July 2013

History

Total project funding amount for 7 projects is $6,875,321*

Export

** Only NIH, CDC and FDA funding data*

Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
3P41GM103694-06S1		CUMMINGS, RICHARD D	BETH ISRAEL DEACONESS MEDICAL CENTER	2018	NIGMS	NIGMS	$1,163,832
National Center for Functional Glycomics							
5P41GM103694-06		CUMMINGS, RICHARD D	BETH ISRAEL DEACONESS MEDICAL CENTER	2017	NIGMS	NIGMS	$1,150,531
National Center for Functional Glycomics							
5P41GM103694-05		CUMMINGS, RICHARD D	BETH ISRAEL DEACONESS MEDICAL CENTER	2016	NIGMS	NIGMS	$1,165,555
National Center for Functional Glycomics							
7P41GM103694-04		CUMMINGS, RICHARD D	BETH ISRAEL DEACONESS MEDICAL CENTER	2015	NIGMS	NIGMS	$800,606
National Center for Functional Glycomics							

Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
5P41GM103694-03		CUMMINGS, RICHARD D	EMORY UNIVERSITY	2015	NIGMS	NIGMS	$195,688
National Center for Functional Glycomics							
5P41GM103694-02		CUMMINGS, RICHARD D	EMORY UNIVERSITY	2014	NIGMS	NIGMS	$1,088,787
National Center for Functional Glycomics							
1P41GM103694-01		CUMMINGS, RICHARD D	EMORY UNIVERSITY	2013	NIGMS	NIGMS	$1,310,322

Similar Projects

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
205	5R21AI156304-02		ABEDON, STEPHEN T	OHIO STATE UNIVERSITY	2022	NIAID	NIAID	$234,000
MALDI imaging of glial scar-forming glycans in Alzheimers disease								
205	5R21AG074152-02		ALONGE, KIMBERLY MICHELE	UNIVERSITY OF WASHINGTON	2022	NIA	NIA	$220,625
Metal Homeostasis in Yeast								
92	5R01GM041840-32		ALSPAUGH, ANDREW	DUKE UNIVERSITY	2022	NIGMS	NIGMS	$315,618
Activated Sugars, Oligosaccharides and Glycans for Antiviral Research								
205	5R44GM140755-02		AMINOVA, LEILA	MIDWEST BIOPROCESSING CENTER, LLC	2022	NIGMS	NIGMS	$868,039
Investigating a Novel Co-regulation of Multi-Drug Efflux Pumps and Polysaccharide Capsule in E. coli								
205	5K08AI123524-05		ARSHAD, MEHREEN	LURIE CHILDREN'S HOSPITAL OF CHICAGO	2021	NIAID	NIAID	$188,478
Assessment of Clinical Grade Near-Native Cleavage-Independent Trimer Protein Compared to Membrane-Bound mRNA Expression Coupled with Glycan Deletion, Restoration and Heterologous Boosting								
205	75N93020D00007-P00002-759302100002-1		AYE, PYONE	TULANE UNIVERSITY OF LOUISIANA	2022	NIAID	NIAID	$262,752
Self-Propagating Mechanism of Prion Diseases								
92	5R01NS045585-20		BASKAKOV, ILIA V	UNIVERSITY OF MARYLAND BALTIMORE	2022	NINDS	NINDS	$508,003
Sialylation-dependent mechanisms driving pancreatic cancer progression								
103	5U01CA233581-05		BELLIS, SUSAN L WELLS, LANCE	UNIVERSITY OF ALABAMA AT BIRMINGHAM	2022	NCI	NCI	$503,659
Using siglecs and their ligands to treat allergic diseases SALTAD								
127	5U19AI136443-05		BOCHNER, BRUCE S	NORTHWESTERN UNIVERSITY AT CHICAGO	2022	NIAID	NIAID	$1,505,741
Immune modulation during infection by a secreted fungal polysaccharide								
205	5R01AI130248-05		BROWN, JESSICA CONRAD	UNIVERSITY OF UTAH	2022	NIAID	NIAID	$381,250
Development of the GlycoHCCTyper for the early detection of HCC								

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
217	1R41CA268268-01		CASTELLINO, STEPHEN MEHTA, ANAND S.	GLYCOPATH, INC	2022	NCI	NCI	$428,443
The role of arginine catabolic pathways in modulating staphylococcal fitness								
98	5R01AI125588-05		CHITTEZHAM THOMAS, VINAI	UNIVERSITY OF NEBRASKA MEDICAL CENTER	2022	NIAID	NIAID	$376,250
Polyclonality of carbapenem resistant Enterobacteriaceae bloodstream infections								
88	5R21AI152018-02		CLANCY, CORNELIUS J	UNIVERSITY OF PITTSBURGH AT PITTSBURGH	2022	NIAID	NIAID	$204,571
N-glycosylation and Immunotherapy for cancer								
228	5U01CA233078-05		DEMETRIOU, MICHAEL	UNIVERSITY OF CALIFORNIA-IRVINE	2022	NCI	NCI	$673,731
Cancer Immunotherapy Targeting Tn Antigen								
113	1R41CA261408-01A1		DEMETRIOU, MICHAEL ZHOU, RAYMOND WENHOU	GLYTR THERAPEUTICS, INC.	2021	NCI	NCI	$398,178
CRYPTOCOCCAL GLYCAN SYNTHESIS								
205	5R01AI135012-05		DOERING, TAMARA L	WASHINGTON UNIVERSITY	2022	NIAID	NIAID	$470,602
Deciphering Helicobacter pylori's glycocode: uncovering and harnessing drug targets								
217	2R15GM109397-03		DUBE, DANIELLE H.	BOWDOIN COLLEGE	2021	NIGMS	NIGMS	$400,669
Novel carbohydrate binding functions of the CR3 I-domain modulate gonococcal-cervical cell interactions								
165	5R01AI134848-05		EDWARDS, JENNIFER L JENNINGS, MICHAEL PAUL	RESEARCH INST NATIONWIDE CHILDREN'S HOSP	2022	NIAID	NIAID	$387,535
Functional Annotation of Glycan Motifs using Common-Fund Data Resources								
240	1R03OD034495-01		EDWARDS, NATHAN J	GEORGETOWN UNIVERSITY	2022	OD	NIDCR OD	$1 $311,999
UCSD Biomedical Scientist Career Development Program in Glycoscience								
205	5K12HL141956-05		ESKO, JEFFREY D GODULA, KAMIL VARKI, AJIT P	UNIVERSITY OF CALIFORNIA, SAN DIEGO	2022	NHLBI	NHLBI	$1,050,483
Analysis of the Initiation of an HIV Broadly Neutralizing Antibody Lineage in a Single Host								
100	1R15AI150484-01A1		FERA, DANIELA	SWARTHMORE COLLEGE	2020	NIAID	NIAID	$329,233
New Congenital Disorders of Glycosylation: Therapy and Models								
150	5R01DK099551-09		FREEZE, HUDSON H.	SANFORD BURNHAM PREBYS MEDICAL DISCOVERY INSTITUTE	2022	NIDDK	NIDDK	$546,107
Functional role of miR-511-3p in allergic asthma and its underlying mechanisms								
100	5R01AI153331-02		GAO, PEISONG	JOHNS HOPKINS UNIVERSITY	2022	NIAID	NIAID	$623,098
Development of a Carbohydrate Microarray								

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
146	1ZIABC010675-18		GILDERSLEEVE, JEFFREY	DIVISION OF BASIC SCIENCES - NCI	2022	NCI	NCI	$315,701
Carbohydrate Microarray Profiling of Glycan-Binding Antibodies and Lectins								
276	1ZIABC010740-17		GILDERSLEEVE, JEFFREY	DIVISION OF BASIC SCIENCES - NCI	2022	NCI	NCI	$1,104,956
Development of novel protein-based vaccine formulations to prevent pneumococcal colonization and disease								
127	5R21AI148722-02		GONZALEZ JUARBE, NORBERTO	J. CRAIG VENTER INSTITUTE, INC.	2021	NIAID	NIAID	$283,588
Q-band Upgrade to an X-band Pulsed EPR spectrometer								
103	1S10OD028706-01A1		GRIFFIN, ROBERT GUY	MASSACHUSETTS INSTITUTE OF TECHNOLOGY	2021	OD	OD	$920,209
A Biomedical Mass Spectrometry Resource: Ongoing Driving Biomedical Projects								
126	5R24GM136766-03		GROSS, MICHAEL L	WASHINGTON UNIVERSITY	2022	NIGMS	NIGMS	$659,894
Targeted disruption the enzymes of O-GlcNAc cycling: Animal models of Disease								
92	1ZIADK060101-16		HANOVER, JOHN A.	NATIONAL INSTITUTE OF DIABETES AND DIGESTIVE AND KIDNEY DISEASES	2022	NIDDK	NIDDK	$1,664,637
Enzymes of O-GlcNAc cycling linked to type-2 diabetes and neurodegeneration								
107	1ZIADK060103-16		HANOVER, JOHN A.	NATIONAL INSTITUTE OF DIABETES AND DIGESTIVE AND KIDNEY DISEASES	2022	NIDDK	NIDDK	$1,358,887
Towards a New Generation of Glycoengineered Pneumococcal Bioconjugate Vaccines								
133	5R44AI131742-04		HARDING, CHRISTIAN	VAXNEWMO, LLC	2022	NIAID	NIAID	$984,934
Messenger RNA immunogens for initiation of HIV V3-glycan neutralizing B cell lineages								
205	5U19AI135902-05		HAYNES, BARTON F.	DUKE UNIVERSITY	2022	NIAID	NIAID	$6,183,078
Glycans in Blood Homeostasis and Disease								
205	5K12HL141954-05		HOFFMEISTER, KARIN MARIA DAHMS, NANCY M. DESAI, UMESH RAMANLAL LAU, JOSEPH TY	VERSITI WISCONSIN, INC.	2022	NHLBI	NHLBI	$972,000
Evolving New Glycosaminoglycan Mimetics								
91	5R01GM127920-04		HSIEH-WILSON, LINDA C	CALIFORNIA INSTITUTE OF TECHNOLOGY	2021	NIGMS	NIGMS	$384,183
Ciliary pcoket matrix in photoreceptor health								
129	5R01EY028555-05		HU, HUAIYU	UPSTATE MEDICAL UNIVERSITY	2022	NEI	NEI	$392,850
Brain Recovery after Cardiac Arrest with Metabolic Glycoengineered Stem Cells								
120	5R01NS110387-05		JIA, XIAOFENG	UNIVERSITY OF MARYLAND BALTIMORE	2022	NINDS	NINDS	$337,969
Enzymatic Synthesis of Sugar-Derived Biosurfactants Using Multifunctional Ionic Liquids								

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
92	1R15GM143682-01		KIM, SUNG JOON	HOWARD UNIVERSITY	2021	NIGMS	NIGMS	$448,706
Defining a novel strategy of adhesion to host carbohydrates employed by Streptococcus oralis								
90	5R21AI149414-02		KING, SAMANTHA JANE	RESEARCH INST NATIONWIDE CHILDREN'S HOSP	2021	NIAID	NIAID	$223,776
Glycoantigen evolution targeting antibodies of the PGT 128 lineage								
107	5R01AI090745-08		KRAUSS, ISAAC JONATHAN	BRANDEIS UNIVERSITY	2022	NIAID	NIAID	$525,019
Mechanisms Of Humoral Evasion								
107	1ZIAAI005023-21		KWONG, PETER D	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$1,269,871
Characterization of a novel S. aureus biofilm polysaccharide								
205	5R21AI159212-02		LEE, CHIA Y.	UNIV OF ARKANSAS FOR MED SCIS	2022	NIAID	NIAID	$228,000
Roles of sialidase in G. vaginalis-host interactions during bacterial vaginosis								
119	5R01AI114635-08		LEWIS, AMANDA L. LEWIS, WARREN G	UNIVERSITY OF CALIFORNIA, SAN DIEGO	2022	NIAID	NIAID	$585,517
High-Throughput De Novo Glycan Sequencing								
205	5R01GM132675-04		LIN, CHENG HONG, PENGYU	BOSTON UNIVERSITY MEDICAL CAMPUS	2022	NIGMS	NIGMS	$447,308
Structure-Function Studies of the HIV-1 Envelope Glycoproteins								
92	1ZIAAI001197-09		LUSSO, PAOLO	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$1,549,551
Optimization and preclinical development of a TB Multiple Antigen Presenting System (MAPS) vaccine								
191	5R01AI135720-05		MALLEY, RICHARD	BOSTON CHILDREN'S HOSPITAL	2022	NIAID	NIAID	$1,144,476
National Center for In Vivo Metabolism								
97	5P41EB015908-33		MALLOY, CRAIG R	UT SOUTHWESTERN MEDICAL CENTER	2021	NIBIB	NIBIB	$1,190,238
Hopping Sugars: molecular mechanism of protonatedglycans rearrangement								
205	5SC2GM135145-03		MARIANSKI, MATEUSZ RYSZARD	HUNTER COLLEGE	2022	NIGMS	NIGMS	$156,000
The role of metal ion homeostasis in regulating bacterial capsule production								
205	1R15AI149725-01A1		MARTIN, JULIA E.	IDAHO STATE UNIVERSITY	2022	NIAID NIGMS	NIAID NIGMS	$88,497 $320,000
The Biology of HIV Infections								
150	1ZIAAI001213-07		MARTIN, MALCOLM	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$2,004,520
Total plasma and IgG glycomes, statin therapy and ASCVD events								
205	5R01HL117861-08		MORA, SAMIA	BRIGHAM AND WOMEN'S HOSPITAL	2022	NHLBI	NHLBI	$447,500
Origin of N-Glycan Site-Specific Heterogeneity								

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
205	5R01GM130915-04		MOREMEN, KELLEY W. KANNAN, NATARAJAN WELLS, LANCE WOOD, ZACHARY ARTHUR	UNIVERSITY OF GEORGIA	2022	NIGMS	NIGMS	$845,521
Core Fucosylation of N-linked Glycans as a Regulator of CD8+ T cell Activation and Function								
217	5R21AI159401-02		NOLZ, JEFFREY C.	OREGON HEALTH & SCIENCE UNIVERSITY	2022	NIAID	NIAID	$192,500
Structure-Based Designs of HIV-1 Immunogens Targeting Germline Precursors of Broadly Neutralizing Antibodies								
172	6R21AI157799-02		PANCERA, MARIE	FRED HUTCHINSON CANCER CENTER	2022	NIAID	NIAID	$220,000
Relevance of serum mannosidase resistance to the specificity of oligomannose-targeting antibodies								
100	5R21AI157317-02		PANTOPHLET, RALPH ALPHONSO	SIMON FRASER UNIVERSITY	2022	NIAID	NIAID	$171,731
Engineering of glycosyltransferases to obtain glycan binding proteins								
298	5R21GM139160-02		PARK, SHELDON NEELAMEGHAM, SRIRAM	STATE UNIVERSITY OF NEW YORK AT BUFFALO	2021	NIGMS	NIGMS	$194,606
Quorum: glycopeptide discovery via mass spectrometry and spectral networking								
201	5R44GM125485-04		PATEL, ANAND	ABTERRA BIOSCIENCES, INC.	2021	NIGMS	NIGMS	$675,037
Expanding the Glycomics toolbox: Chemo-enzymatic synthesis of well-defined N-glycan standards								
205	5R43GM137724-02		PRUDDEN, ANTHONY	VIAMUNE, INC.	2021	NIGMS	NIGMS	$274,486
Lysosomal Diseases								
119	1ZIAHL000140-18		PUERTOLLANO, ROSA	NATIONAL HEART, LUNG, AND BLOOD INSTITUTE	2022	NHLBI	NHLBI	$699,963
Research Project 3 Galectin-1-glycan interactions: Novel regulatory checkpoints linking immunosuppression and angiogenesis in virally induced cancers								
236	5U54CA221208-06	6412	RABINOVICH, GABRIEL	UNIVERSITY OF MIAMI SCHOOL OF MEDICINE	2022	NCI	NCI	*$73,541*
Positive Selection as a Host Mechanism to Control the Microbiome								
153	1DP2GM136652-01		RAKOFF-NAHOUM, SETH	BOSTON CHILDREN'S HOSPITAL	2019	NIGMS	OD	$2,655,000
UM CFAR SCCC Argentina Consortium for research and training in Virally Induced AIDS Malignancies								
103	5U54CA221208-06		RAMOS, JUAN CARLOS COSO, OMAR ADRIAN	UNIVERSITY OF MIAMI SCHOOL OF MEDICINE	2022	NCI	NCI	$488,428
Mucin Glycans in the Regulation of Microbial Virulence								
205	5R01EB017755-08		RIBBECK, KATHARINA	MASSACHUSETTS INSTITUTE OF TECHNOLOGY	2022	NIBIB	NIBIB	$568,885
Role of a UPP-like exopolysaccharide in Brucella virulence								
205	5R21AI148132-02		ROOP, ROY M	EAST CAROLINA UNIVERSITY	2021	NIAID	NIAID	$218,927
Structural and functional studies of microbial O-glycosyltransferases that synthesize virulence factors								

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
205	1ZIADE000761-02		SAMARA, NADINE	NATIONAL INSTITUTE OF DENTAL & CRANIOFACIAL RESEARCH	2022	NIDCR	NIDCR	$338,323
Glycoengineering of Therapeutic Peptides for Improved Treatment of Human Diseases								
129	5R01EB025892-04		SAMMAKIA, TAREK	UNIVERSITY OF COLORADO	2021	NIBIB	NIBIB	$333,520
Glycosylation Analysis by Sheath-Flow SERS								
156	5R21GM140382-02		SCHULTZ, ZACHARY	OHIO STATE UNIVERSITY	2022	NIGMS	NIGMS	$192,013
Host proteins that interact with the BCG cell envelope								
122	5R21AI163949-02		SIEGRIST, MARY SLOAN; GRIMES, CATHERINE LEIMKUHLER	UNIVERSITY OF MASSACHUSETTS AMHERST	2022	NIAID	NIAID	$218,307
Natural Glycans for Functional Glycomics								
205	5R44GM133252-03		SMITH, DAVID FLETCHER	NATGLYCAN, LLC	2021	NIGMS	NIGMS	$780,222
A Human Salivary Glycome Discovery Platform for Interrogating Glycan Function in Oral Innate Immunity								
239	1R43DE031990-01		SMITH, DAVID FLETCHER	NATGLYCAN, LLC	2022	NIDCR	NIDCR	$314,971
A Research Resource for Ultra-sensitive High Throughput Proteomics								
143	5P41GM103493-20		SMITH, RICHARD D	BATTELLE PACIFIC NORTHWEST LABORATORIES	2022	NIGMS	NIGMS	$1,593,314
Impact of Diet on Intestinal Microbiota-Host Dynamics								
86	5R01DK085025-13		SONNENBURG, JUSTIN L	STANFORD UNIVERSITY	2022	NIDDK	NIDDK	$353,250
Molecular alterations driving the development of primary prostate cancers								
88	1ZIABC011679-07		SOWALSKY, ADAM	DIVISION OF BASIC SCIENCES - NCI	2022	NCI	NCI	$777,497
Lectin Resistant Tumor Cells and Functional Glycomics								
277	5R01GM105399-36		STANLEY, PAMELA M	ALBERT EINSTEIN COLLEGE OF MEDICINE	2022	NIGMS	NIGMS	$447,991
Construction of chondroitin sulfate microarray								
132	1R43GM140693-01A1		SU, GUOWEI	GLYCAN THERAPEUTICS CORPORATION	2022	NIGMS	NIGMS	$331,634
Inhibition of coronavirus release and mitigate COVID-19 pathogenesis								
144	1ZIAAI001263-03		SUN, PETER D.	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$171,318
Activation of Fc receptors by antibody and innate acute phase proteins								
135	1ZIAAI000853-24		SUN, PETER D.	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$341,728
Recognition Of HIV Envelope gp120 by Siglec and C-type Lectin Receptors								
90	1ZIAAI000880-22		SUN, PETER D.	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$683,451
BioCARS: Structural Dynamics and Biological Mechanisms								

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
187	5P41GM118217-03		TIRRELL, MATTHEW MOFFAT, JOHN KEITH RANGANATHAN, RAMA SRAJER, VUKICA	UNIVERSITY OF CHICAGO	2021	NIGMS	NIGMS	$1,289,704
Laboratory of Neuro Imaging Resource (LONIR)								
86	5P41EB015922-25		TOGA, ARTHUR W THOMPSON, PAUL M	UNIVERSITY OF SOUTHERN CALIFORNIA	2022	NIBIB	NIBIB	$1,217,435
Role and Mechanisms of E4BP4 over-activation in diet-induced fatty liver disease								
90	5R01DK121170-04		TONG, XIN	UNIVERSITY OF MICHIGAN AT ANN ARBOR	2022	NIDDK	NIDDK	$411,832
Biology of Sialic Acids and their Substitutions								
123	5R01GM032373-38		VARKI, AJIT P	UNIVERSITY OF CALIFORNIA, SAN DIEGO	2022	NIGMS	NIGMS	$521,422
Heparan Sulfate 3-O-Sulfation in Transcellular Propagation of Tauopathy in Alzheimer's Disease								
113	1RF1AG069039-01		WANG, CHUNYU LINHARDT, ROBERT J LIU, JIAN WANG, LIANCHUN	RENSSELAER POLYTECHNIC INSTITUTE	2020	NIA	NIA	$2,595,120
Convergent Chemoenzymatic Synthesis of Glycopeptides and Glycoproteins								
150	5R01GM080374-13		WANG, LAI-XI	UNIV OF MARYLAND, COLLEGE PARK	2021	NIGMS	NIGMS	$307,955
A Virion-Display Oscillator Array and Detection Platform for Quantification of Transmembrane Protein Binding Kinetics								
94	5R33CA235294-03		WANG, SHAOPENG	ARIZONA STATE UNIVERSITY-TEMPE CAMPUS	2022	NCI	NCI	$362,612
Peptidoglycan binding protein specificity and bacterial cell division								
235	5R01GM125656-04		WEISS, DAVID S	UNIVERSITY OF IOWA	2021	NIGMS	NIGMS	$328,751
Neuroimaging Analysis Center (NAC)								
171	5P41EB015902-25		WESTIN, CARL-FREDRIK KIKINIS, RON	BRIGHAM AND WOMEN'S HOSPITAL	2022	NIBIB	NIBIB	$1,324,073
Transitioning GLYCAM-Web to a self-sustaining carbohydrate modeling service								
98	5R24GM136984-03		WOODS, ROBERT J	UNIVERSITY OF GEORGIA	2022	NIGMS	NIGMS	$415,250
Computational tools to aid the design of glycomimetic agents								
141	5R01GM135473-03		WOODS, ROBERT J	UNIVERSITY OF GEORGIA	2022	NIGMS	NIGMS	$374,287
Design of hetero-multivalent targeted liposome for combating intracellular Mycobacterium tuberculosis								
125	5R21AI149383-02		WU, HUNG-JEN CIRILLO, JEFFREY D.	TEXAS ENGINEERING EXPERIMENT STATION	2021	NIAID	NIAID	$227,250
Tumor immune and glycan biomarkers for progressive prostate cancer								

Match Score	Project Number	Sub	Principal Investigator(s)/ Project Leader(s)	Organization	Fiscal Year	Admin IC	Funding IC	FY Total Cost by IC
205	5R01CA212409-05		WU, JENNIFER D ABDULKADIR, SARKI A. DRAKE, RICHARD R.	NORTHWESTERN UNIVERSITY AT CHICAGO	2022	NCI	NCI	$552,871
Pan-specific tools and reagents for the enrichment of sialo-glycans								
228	3R44OD024964-03S1		WU, SHENG-CHENG	GLYCOSENSORS AND DIAGNOSTICS, LLC	2022	OD	OD	$37,200
Center for the Investigation of Factor VIII Inhibitors and Glycosylation								
112	5U54HL142019-05		XIAO, WEIDONG LI, LEI MIAO, CAROL H	INDIANA UNIV-PURDUE UNIV AT INDIANAPOLIS	2022	NHLBI	NHLBI	$1,323,844
Development of streamlined chemoenzymatic glycan remodeling systems for antibodies and other important glycoproteins								
216	5R44GM134816-04		YANG, QIANG	GLYCOT THERAPEUTICS, LLC	2022	NIGMS	NIGMS	$713,666
Mechanism and Anti-Cancer Activity of SCFA-Hexosamine Analogs								
106	5R01CA112314-14		YAREMA, KEVIN J	JOHNS HOPKINS UNIVERSITY	2022	NCI	NCI	$343,015
Immersive Training in Glycosciences								
96	5K12HL141952-05		ZACHARA, NATASHA ELIZABETH HASCALL, VINCENT CHARLES SCHNAAR, RONALD L	JOHNS HOPKINS UNIVERSITY	2022	NHLBI	NHLBI	$954,863
Laboratory Studies of Abnormal Host Defense and Immunoregulatory Diseases								
92	1ZIAAI001151-12		ZAREMBER, KOL	NATIONAL INSTITUTE OF ALLERGY AND INFECTIOUS DISEASES	2022	NIAID	NIAID	$119,145
Glycogen promotes protein aggregation in ALS and FTD								
90	1R21NS127331-01		ZHANG, YONGJIE	MAYO CLINIC JACKSONVILLE	2022	NINDS	NIA	$430,375
New Glycosylation Methods for Microbial Glycan Synthesis								
205	1R15GM147867-01		ZHU, JIANGLONG	UNIVERSITY OF TOLEDO	2022	NIGMS	NIGMS	$451,500
ABC transporter-mediated secretion of capsular polysaccharides								
205	5R21AI164216-02		ZIMMER, JOCHEN	UNIVERSITY OF VIRGINIA	2022	NIAID	NIAID	$197,140
Comprehensive Characterization of Glycosylation Alterations in Alzheimer's Disease								
100	5R01AG062240-05		ZIVKOVIC, ANGELA M JIN, LEE-WAY LEBRILLA, CARLITO B	UNIVERSITY OF CALIFORNIA AT DAVIS	2022	NIA	NIA	$656,160